MIL-S-51078C
25 April 1984
SUPERSEDING
MIL-S-51078B
7 June 1976

MILITARY SPECIFICATION

SODIUM HEXAMETAPHOSPHATE, TECHNICAL

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of sodium hexametaphosphate in various forms suitable for treating potable water.

2.1 Classification. Sodium hexametaphosphate, covered by this specification shall be of the following forms as specified (see 6.2):

Powder
Granular
Bead
Plate

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin
PPP-D-723 - Drums, Fiber
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons)

MILITARY

MIL-B-117 - Bags, Sleeves, and Tubing - Interior Packaging

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1168 - Ammunition Lot Numbering

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

The United States Pharmacopeia, Twentieth Revision

(The United States Pharmacopeia may be obtained from the United States Pharmacopeial Convention, Inc., 12601 Twinbrook Parkway, Rockville, MD 20852.)

ASTM STANDARDS

D 501 - Sampling and Chemical Analysis of Alkaline Detergents
D 1193 - Reagent Water
D 3951 - Commercial Packaging
E 11 - Wire-Cloth Sieves for Testing Purposes

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical properties. Sodium hexametaphosphate shall conform to the chemical properties listed in table I when tested as specified therein.

TABLE I. Chemical properties

Property	Requirement	Test paragraph
Percent phosphate as P_2O_5, minimum	67.0	4.2.4.1
Percent loss on ignition, maximum	1.0	4.2.4.1
Percent water-insoluble matter, maximum	0.05	4.2.4.1
pH of 1-percent aqueous solution at 25°C	6.7 to 7.2	4.2.4.1
Arsenic, ppm as arsenic trioxide, maximum	10.0	4.2.4.2
Heavy metals, ppm as lead, maximum	10.0	4.2.4.3
Additives (sudsing agents)	negative	4.2.4.4

3.2 Particle size.

3.2.1 Powder, granular, and bead. Particle size for powder, granular, and bead forms shall be in accordance with table II when tested as specified in 4.2.4.5.

TABLE II. Particle size

Requirement	Percentage		
	Powder	Granular	Bead
Retained on 4.75-mm sieve	---	---	1.0 (max)
Pass through 2.36-mm sieve	---	98.0 (min)	4.0 (max)
Pass through 600-micrometer sieve	98.0 (min)	50.0 (min)	---
Pass through 150-micrometer sieve	50.0 (min)	20.0 (max)	---

3.2.2 Plate. Plate form shall be irregular shaped particles, shall have an equivalent diameter of 1/2 to 3 inches, and shall be 1/16 to 1/4 inches in thickness.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the sodium hexametaphosphate of one form produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. Sampling for test shall be conducted in accordance with table III. A representative specimen of approximately 500 grams (g) shall be removed from each sample container and placed in a clean, dry container labeled to identify the lot and container from which it was taken.

TABLE III. Sampling for test

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit container, ready for shipment. Sample unit containers shall be examined for the following defects using an AQL of 1.5 percent defective:

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(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Unitization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 Certification. When applicable, the contractor shall certify that the plate form conforms to 3.2.2.

4.2.4 Tests. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Chemical properties. Determine compliance with the requirements of table I in accordance with the test methods in table IV as specified.

TABLE IV. Test methods

Requirement	Test Method
Phosphate as P_2O_5	ASTM D 501, total P_2O_5*
Ingition loss	ASTM D 501, loss on ignition
Water insoluble matter	ASTM D 501, matter insoluble in water
pH	ASTM D 501, pH of 1-percent solution

*Any of the three methods may be used; however, in case of dispute, the preferred method shall be used to determine compliance with the requirement.

4.2.4.2 Arsenic. Determine arsenic in accordance with the silver diethyldithiocarbamate method specified in U.S. Pharmacopeia (see 6.4).

4.2.4.3 Heavy metals.

(a) Standard lead solution. Dissolve 0.0160 g of lead nitrate in water to make 1000 milliliters (mL). One mL of standard lead solution equals 0.01 milligrams of lead.

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(b) Procedure. Dissolve 5.000 g of specimen in 35 mL of water; add 2 drops of phenolphthalein solution (1-percent in ethanol) and 10-percent ammonium hydroxide until a slight pink color is produced. Transfer quantitatively to a 50-mL volumetric flask and make up to volume. Pipet a 30-mL aliquot into one Nessler tube and a 10-mL aliquot into a second matching Nessler tube. Consider the 30-mL aliquot as the specimen and the 10-mL aliquot as the control. Add 2 mL of the standard lead solution to the control. Add 2 mL of 1N acetic acid to each (specimen and control). Add sufficient water to each to make up to 40 mL. Add 10 mL of hydrogen sulfide water to each. After mixing, the color developed by the specimen shall not exceed the color developed by the control.

4.2.4.4 Additives. Shake 1.0 g of specimen in 125 mL of water. If suds stable for 5 minutes are formed, the material fails the test.

4.2.4.5 Particle size (powder, granular, and bead forms, only). Tare and nest the appropriate size sieves conforming to ASTM E 11 in descending order (coarsest sieve on top; finest sieve on bottom) on a tared receiving pan. Accurately weigh 100.0 ± 0.1 g of specimen into the top pan, cover, and shake for 10 minutes on a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the portions retained on the sieves and in the receiving pan and calculate compliance with 3.2.1.

5. PACKAGING

5.1 Unit packing. Unit packing shall be level A, B or industrial as specified (see 6.2).

5.1.1 Level A. A quantity of 100 (+1.0 or -0.0) pounds (lb) of sodium hexametaphosphate shall be unit packed in a steel pail conforming to type II, class 6 of PPP-P-704 furnished with a bag liner conforming to type I, class B, style 2 of MIL-B-117. The bag liner shall be closed by tying, knotting or heat sealing. The pail shall be closed by bolted locking ring as specified in PPP-P-704.

5.1.2 Level B. A quantity of 100 (+1.0 or -0.0) lb of sodium hexametaphosphate shall be unit packed in a fiber drum furnished with a bag liner. The fiber drum shall conform to type II, grade A, class 1 of PPP-D-723, with an innermost lining of polyethylene film conforming to type I, class 2, grade optional, finish 1 of L-P-378 having a minimum thickness of 0.002 inch. The bag liner shall conform to type I, class B, style 2 of MIL-B-117. The bag shall be closed by tying, knotting or heat sealing, and the drum shall be closed as specified in the appendix to PPP-D-723.

5.1.3 Industrial. A quantity of 100 (+1.0 or -0.0) lb of sodium hexametaphosphate shall be unit packed in accordance with ASTM D3951.

5.2 Packing. Sodium hexametaphosphate, unit packed as specified in 5.1, shall require no further protection for shipment other than unitization.

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5.2.1 Unitization. Identical quantities of level A or B packs of sodium hexametaphosphate shall be palletized in accordance with load type III, bonding means E of MIL-STD-147 for metal pails and load type IV of MIL-STD-147 for fiber drums. Industrial packs shall be unitized in a manner to assure protection of the packs from supply source to destination and to assure carrier acceptance.

5.3 Marking. Level A and B packs and pallet loads shall be marked in accordance with MIL-STD-129. Industrial packs shall be marked in accordance with ASTM D3951. Each pack shall be marked to show the lot or batch number and date of manufacture of the sodium hexametaphosphate. In addition, each pack shall be marked: "FOR USE IN POTABLE WATER."

6. NOTES

6.1 Intended use. Sodium hexametaphosphate, covered by this specification, is intended for use in the treatment of potable water in concentrations up to 10 parts per million.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Form of material required, and
(c) Level of unit packing required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Arsenic. See U.S. Pharmacopeia XX, page 907. For the arsine generator described therein, see the arsenic generator figure in ASTM E223.

6.5 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodians:

Army - EA
Air Force - 68

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-51078C	SODIUM HEXAMETAPHOSPHATE, TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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